

Q. 1 (An interesting formula). Let X be a random variable that can take the values $0, 1, 2, 3, \dots$. Prove the following interesting formula:

$$\mathbf{E}(X) = \sum_{k=1}^{\infty} \mathbf{P}\{X \geq k\}.$$

[Hint: try to write the probability as an expectation.]

Q. 2 (A stupid game). You roll one (six-sided, fair) die and flip one (two-sided, fair) coin independently. If the coin comes up heads, you tell me the number on the die. However, if the coin comes up tails, you tell me the number on the die times two.

- What is the expectation of the number you tell me?
- You tell me the number 6. Given only this information, do I think it is more likely that your coin flip came up heads or tails? Why?

Q. 3 (Bag of coins). You have a large bag of coins. Two-thirds of the coins in the bag are fair coins. However, one-third of the coins in the bag are Canadian currency: they are slightly more likely to come up heads (65%) than tails (35%).

- You draw one coin from the bag uniformly at random (that is, you are equally likely to pick any coin in the bag) and flip it twice independently. What is the expected number of times you get heads, given that the coin was fair? What is the expected number of times you get heads, given that the coin was Canadian?
- You draw again one coin from the bag uniformly at random and flip it twice independently. What is the expected number of times you get heads?
- Now it is my turn to draw one coin from the bag uniformly at random and flip it twice independently. I tell you that I flipped exactly 1 tails. Given this information, what is the probability that the coin I drew was Canadian currency?

Q. 4 (Love triangles). A dating website works as follows. There are n people signed up. For every pair of people, the website independently flips a coin that comes up heads with probability p . If the coin comes up heads, the website creates a match between this pair of people. (Being a scam, the site does not actually use any specific information about the people in deciding who to hook up.)

In this setting, it is perfectly possible that the website randomly creates a love triangle: three people who are all mutually hooked up.

- a. What is the expected number of hookups a given individual has at once?
- b. What is the probability that a given individual has more than one hookup?
- c. What is the expected number of love triangles on the entire site?

Q. 5 (Overbooked flights). An airplane seats (at most) 200 passengers. Unfortunately, some people forget the time and end up missing their flight. Each booked passenger fails to show up independently with 2% probability.

Each ticket costs \$500. However, the airline only collects the ticket fee for those passengers who are actually seated on the flight (passengers who miss the plane must be resealed on another flight, so the airline loses the ticket fee for the missed flight). Thus having empty seats costs the airline money. In order not to lose money, the airlines overbook their flights: they actually sell 202 tickets for every 200 passenger plane.

- a. What is the probability that the plane is overbooked? (That is, the probability that more people show up than there are seats on the plane.)
- b. What is the expected amount the airline collects in ticket fees for this flight?
- c. If more than 200 passengers show up, some passengers cannot be seated. For those passengers, the airline must refund the price of the ticket *and* pay a \$300 fine. What is the expected total amount of money the airline makes from operating this flight?